

Binary search tree using linked list

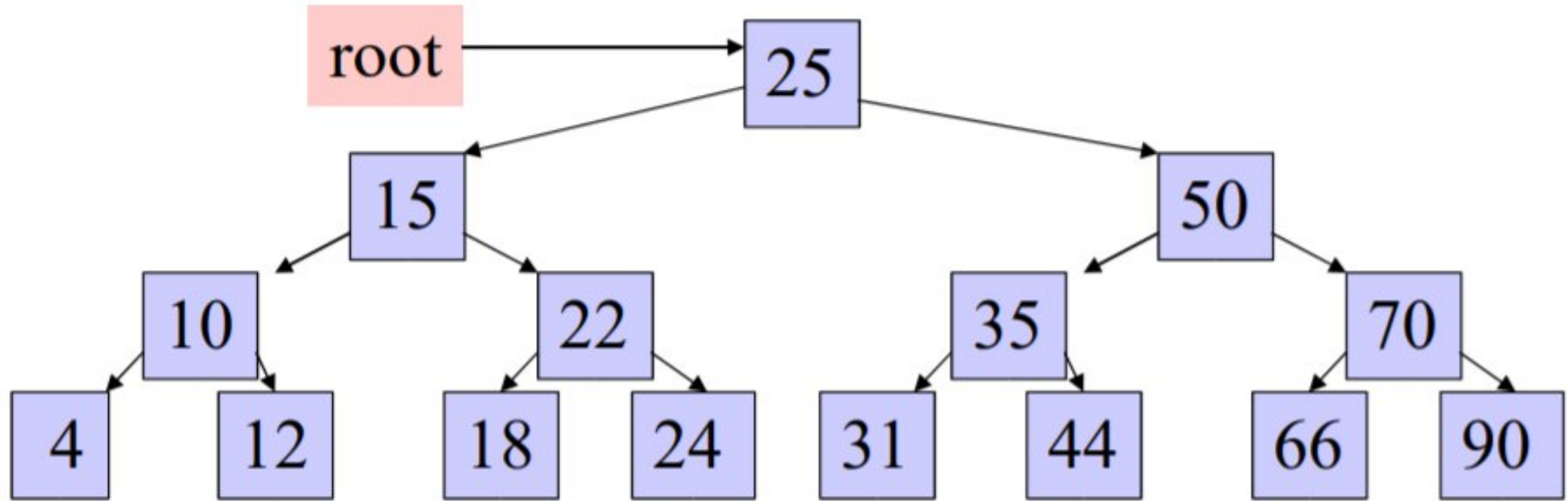
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Traversals (Remind Previous Class)

- Visit every node in the tree and perform some operation on it (ex) print out the data fields of each node
- Three steps to a traversal
 1. Visit the Root node
 2. Traverse its left subtree
 3. Traverse its right subtree
- The order in which you perform these three steps results in different traversal orders:
 - Pre-order traversal: Root node -> Left node -> Right node
 - In-order traversal: left node -> Root node -> Right node
 - Post-order traversal: Left node -> Right node -> Root node

Traversal Examples



Answer

- InOrder (root) visits nodes in the following order:

4, 10, 12, 15, 18, 22, 24, 25, 31, 35, 44, 50, 66, 70, 90

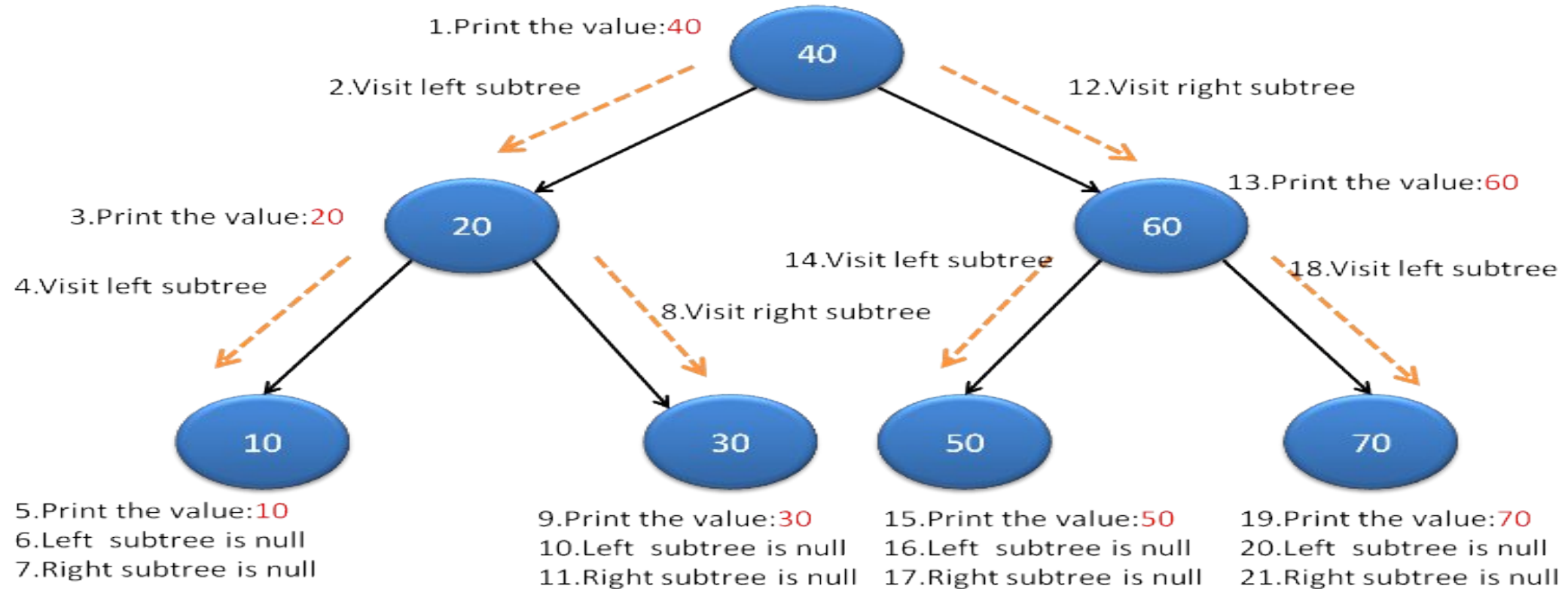
A Pre-order traversal visits nodes in the following order:

25, 15, 10, 4, 12, 22, 18, 24, 50, 35, 31, 44, 70, 66, 90

A Post-order traversal visits nodes in the following order:

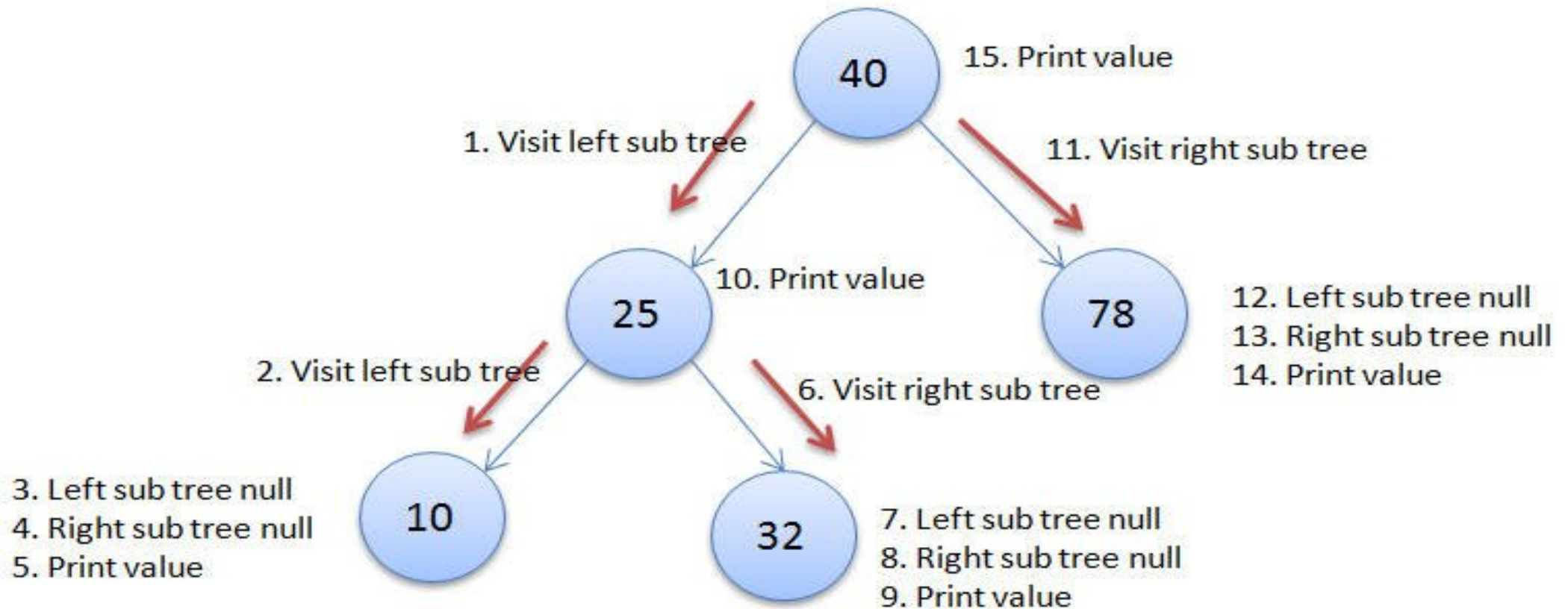
4, 12, 10, 18, 24, 22, 15, 31, 44, 35, 66, 90, 70, 50, 25

Pre-Order :



PreOrder traversal of above graph is : 40, 20, 10, 30, 60, 50, 70

Post-Order:



The above POSTORDER traversal gives: **10, 32, 25, 78, 40**

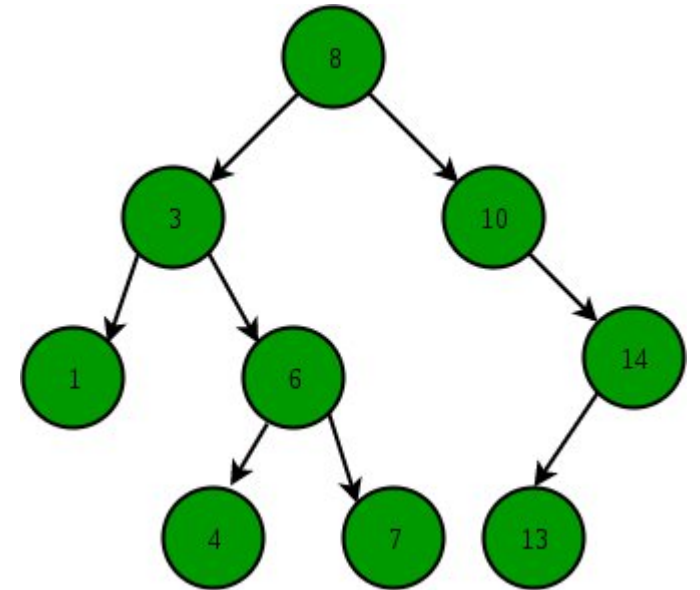
What is Binary search tree?

Binary Search Tree is a node-based binary tree data structure which has the following properties:

=>The left subtree of a node contains only nodes with keys lesser than the node's key.

=>The right subtree of a node contains only nodes with keys greater than the node's key.

=>The left and right subtree each must also be a binary search tree.

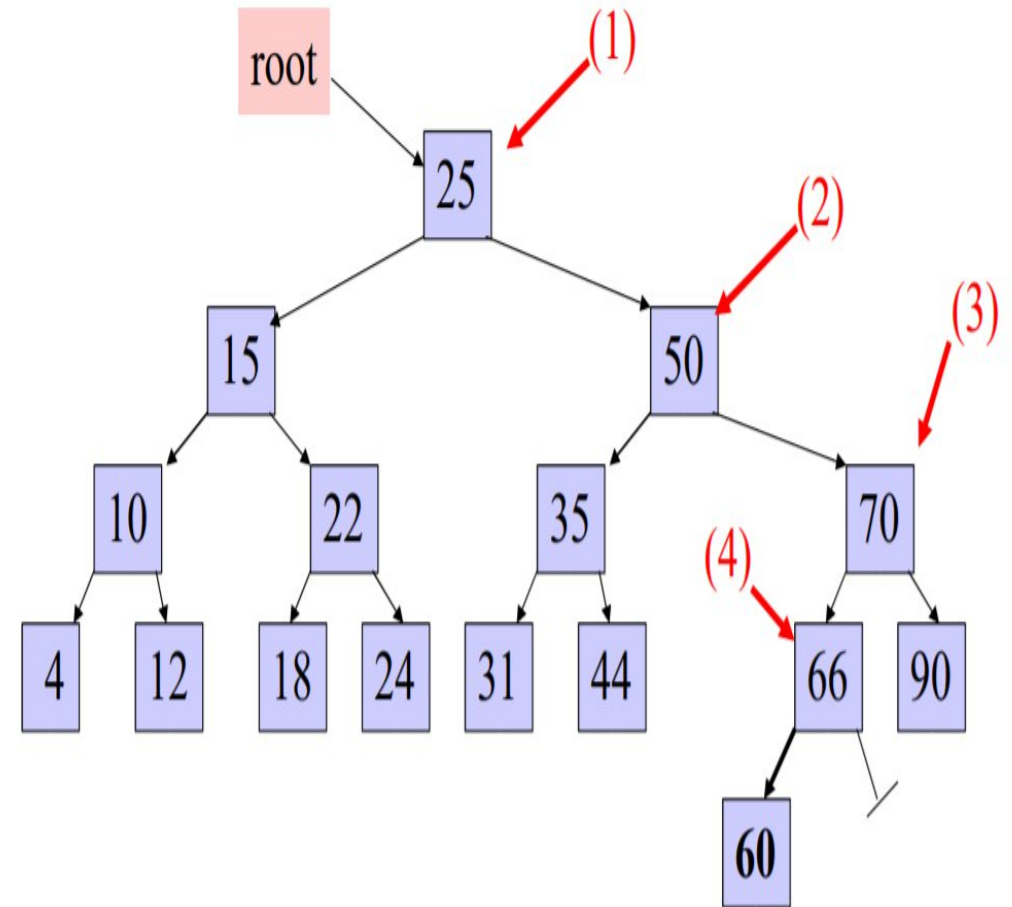


Insertion of Binary search tree

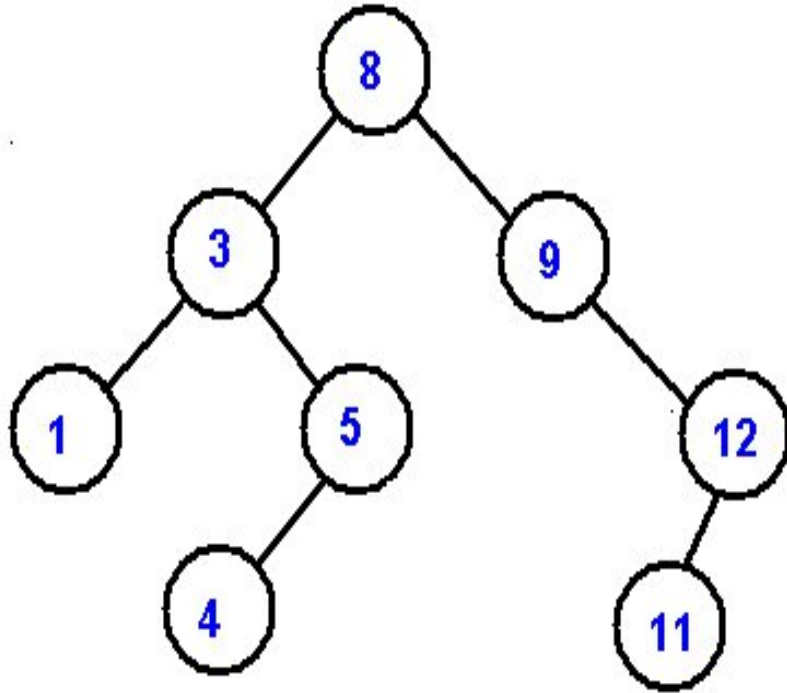
1. Always insert new node as leaf node.
2. Start at root node as current node.
3. If new node's key $<$ current's key.
 1. If current node has a left child, search left.
 2. Else add new node as current's left child.
4. If new node's key $>$ current's key.
 1. If current node has a right child, search right.
 2. Else add new node as current's right child.

Example: insert 60 in the tree

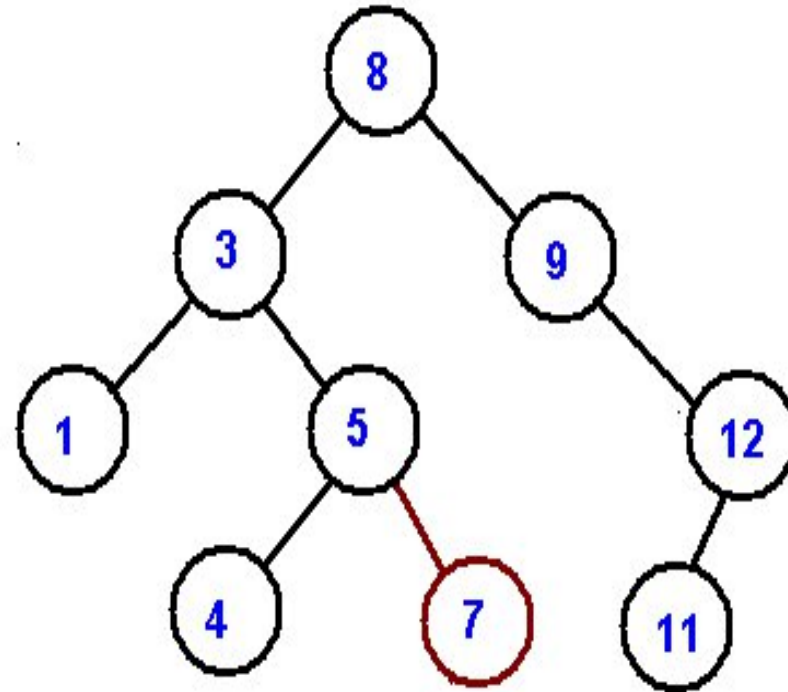
- 1. start at the root, 60 is greater than 25, search in right subtree
- 2. 60 is greater than 50, search in 50's right subtree
- 3. 60 is less than 70, search in 70's left subtree
- 4. 60 is less than 66, add 60 as 66's left child



Insert 7



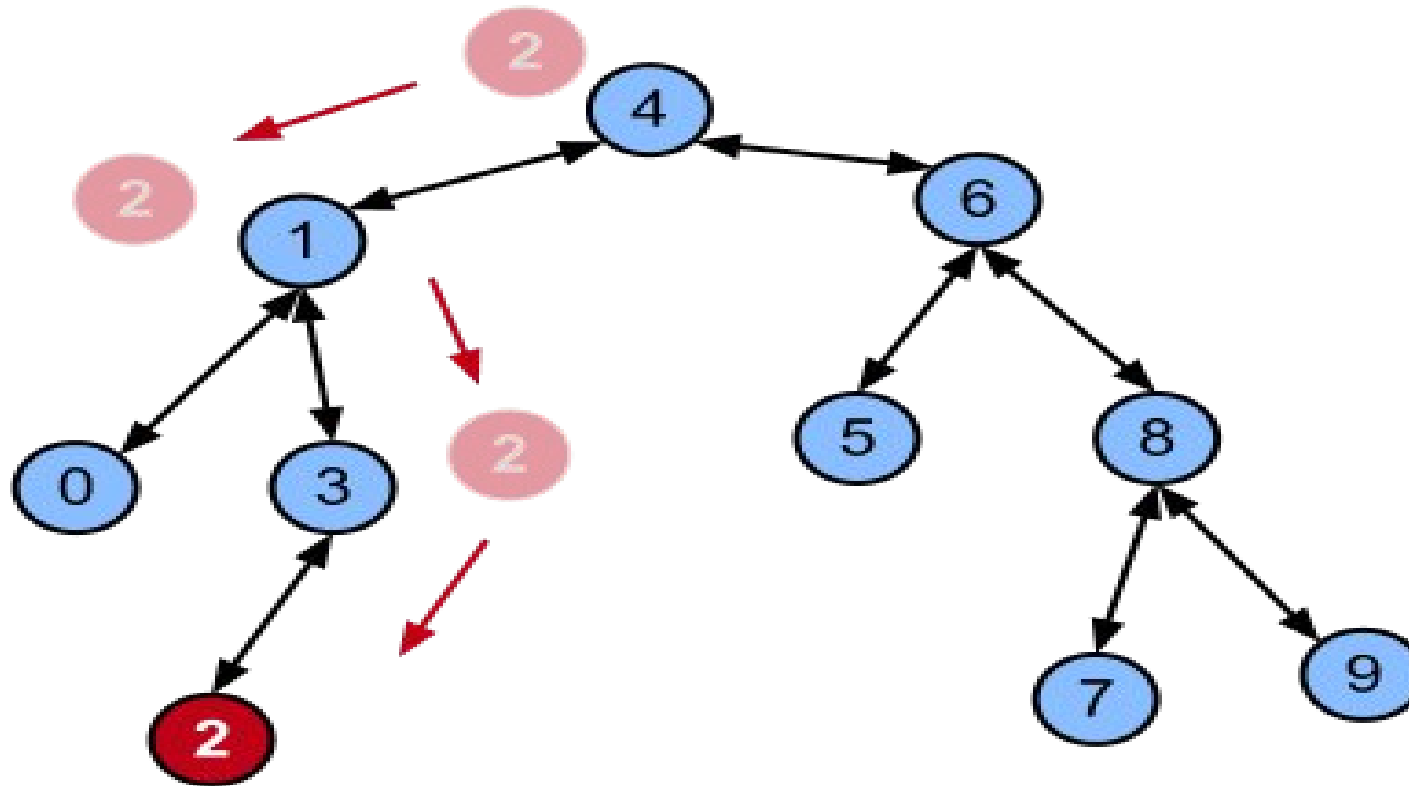
before insertion



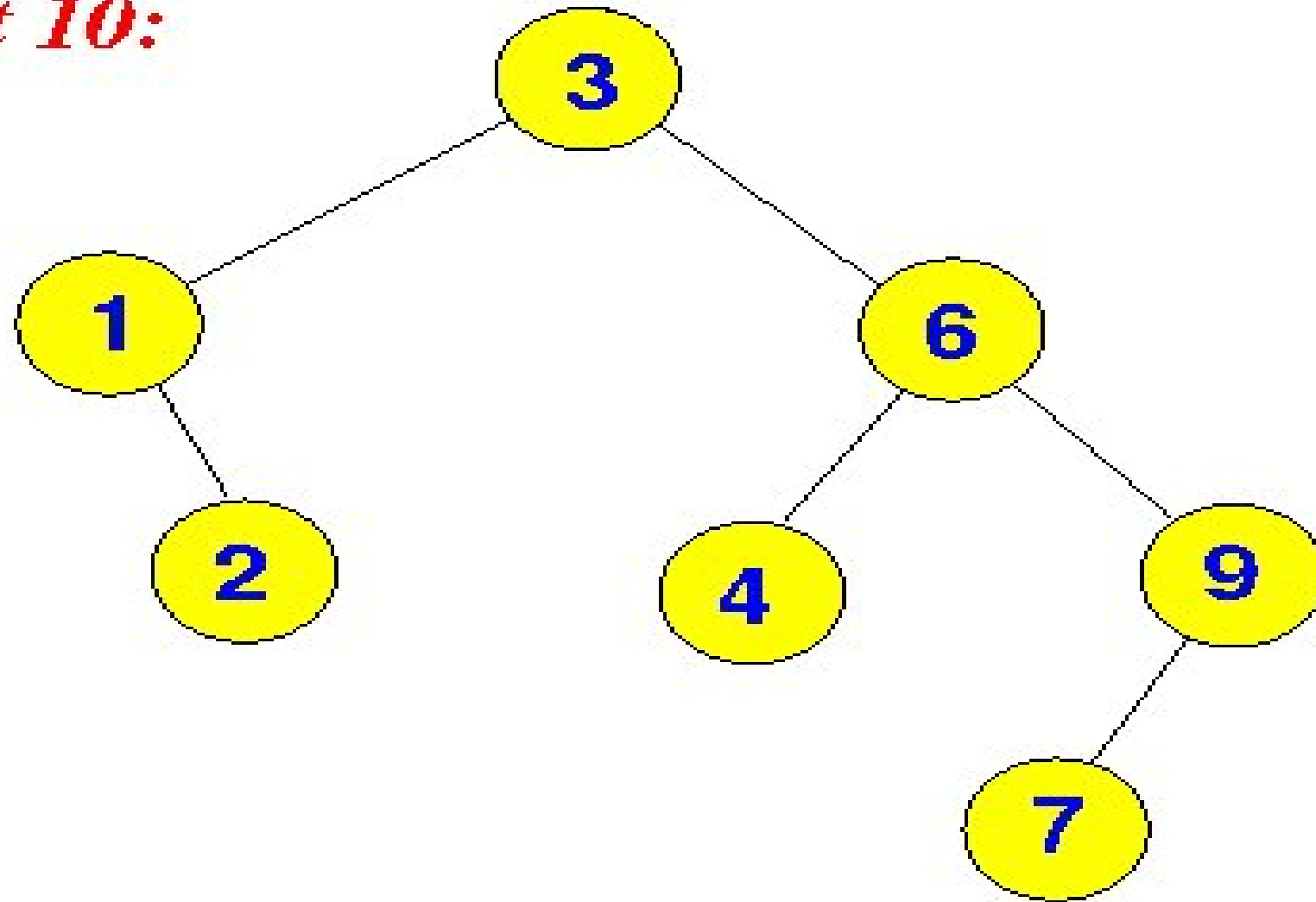
after insertion

Insert 2

Insert in BST



Insert 10:

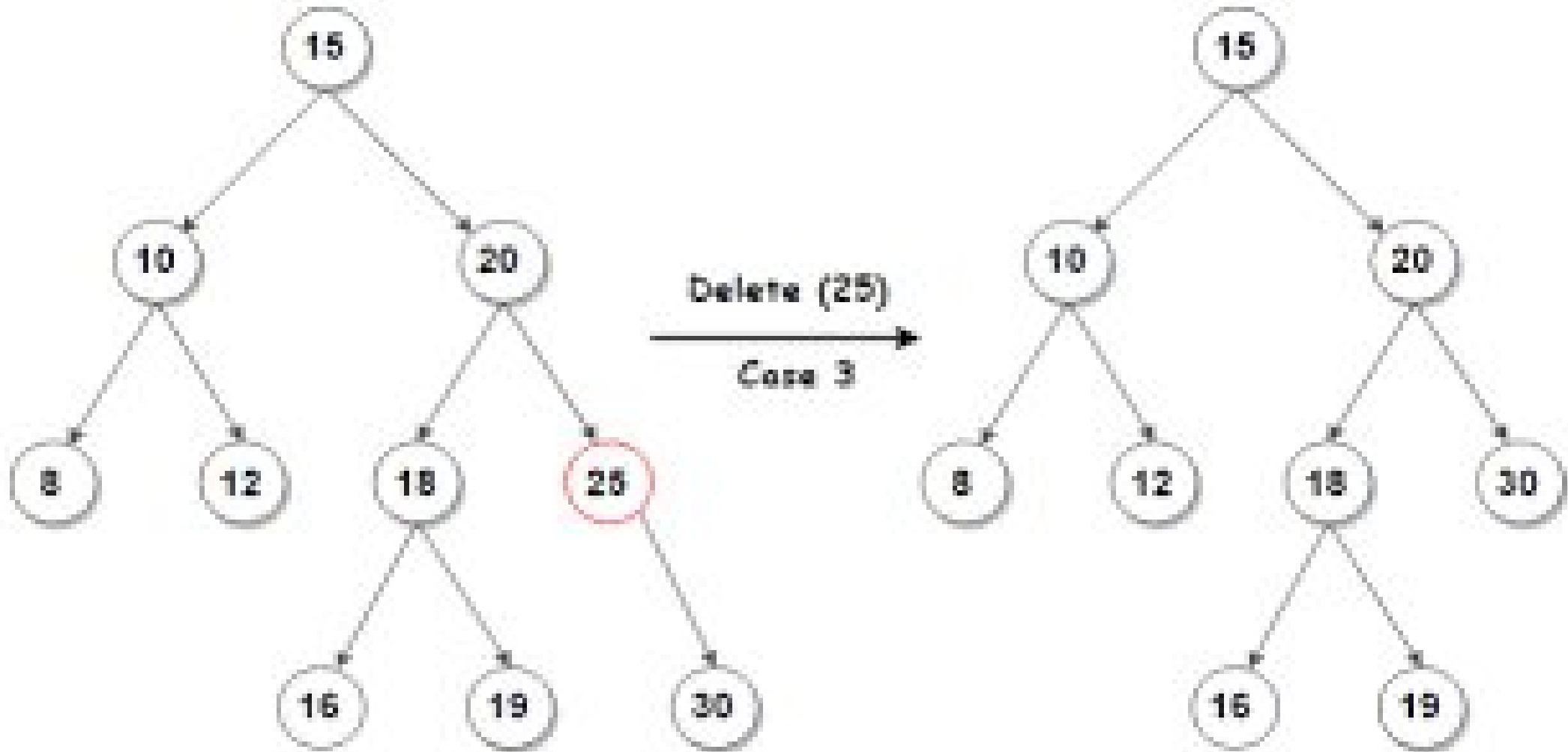


Deletion of Binary search tree

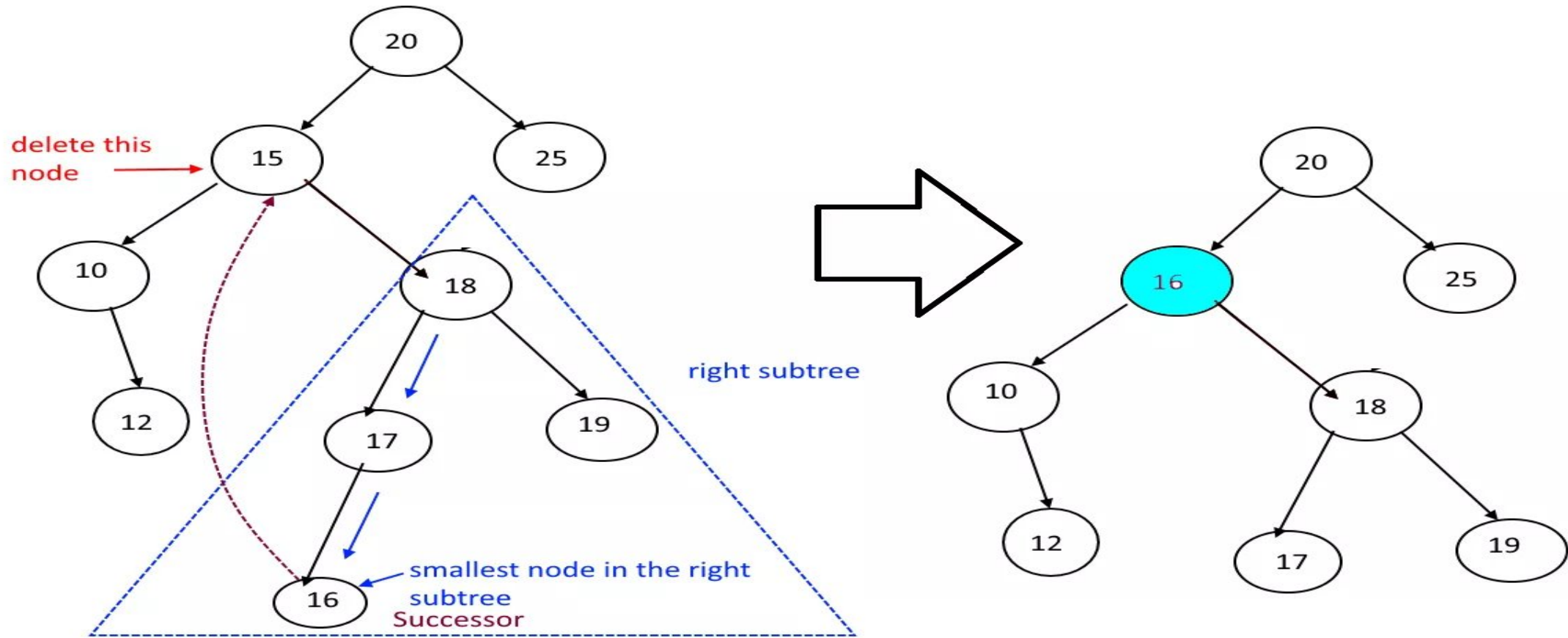
- Find item. Remove node. Repair tree.
- If leaf (no children), delete node.
- If unary (one child), replace by other child.

Described with proper picture in next slide

Delete 25

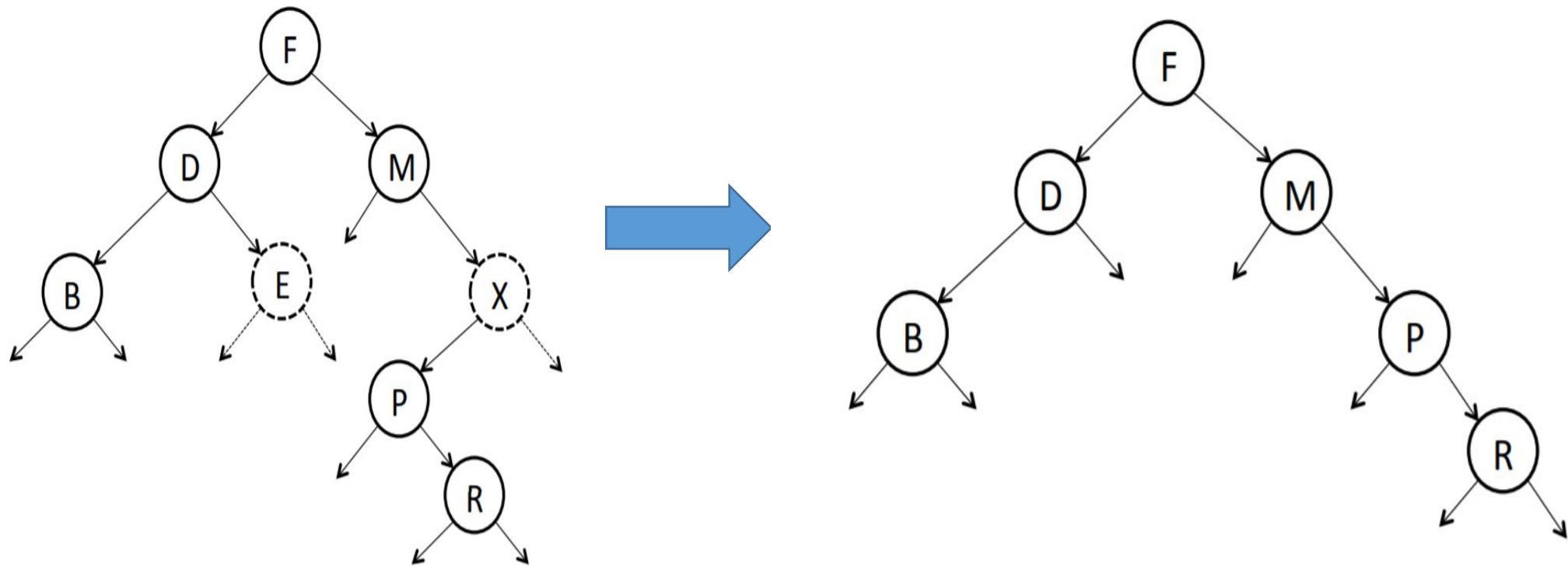


Delete 15



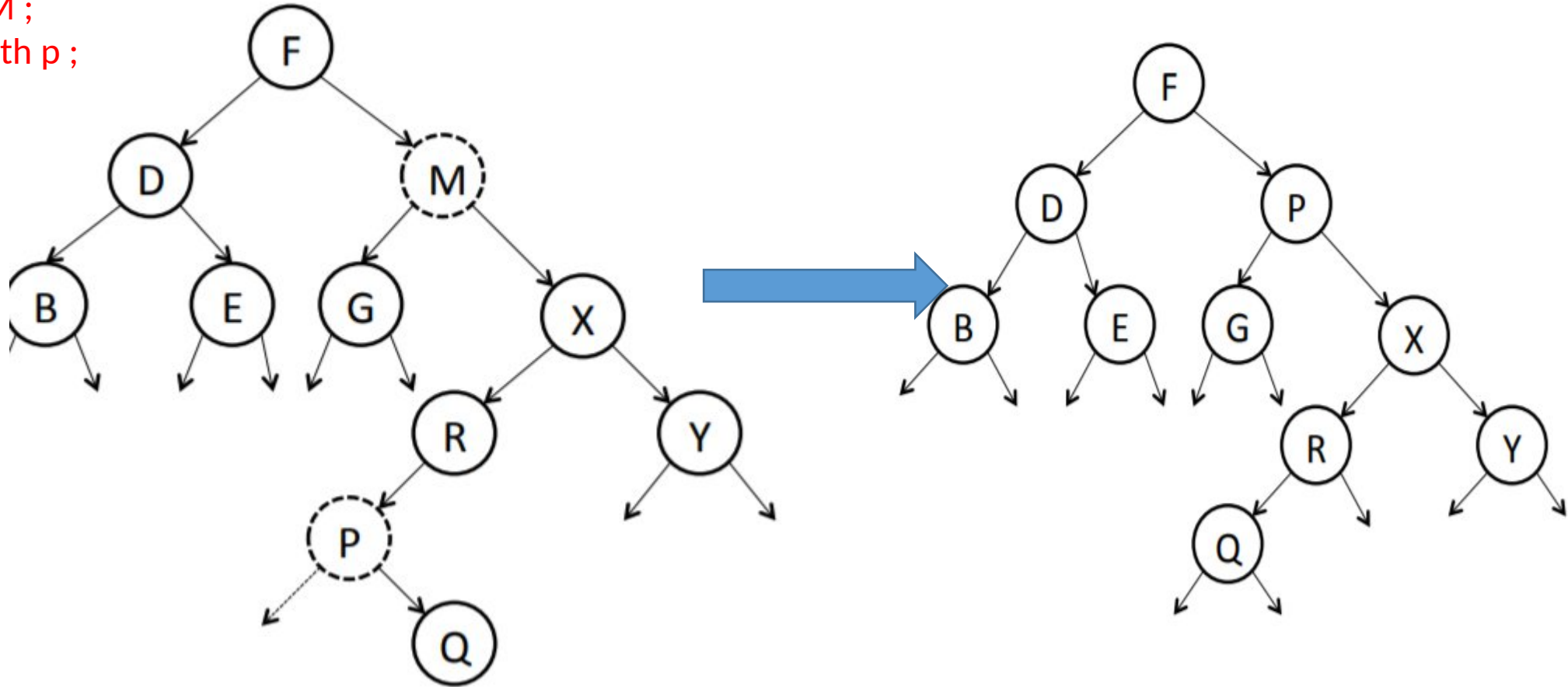
Delete E

Delete X

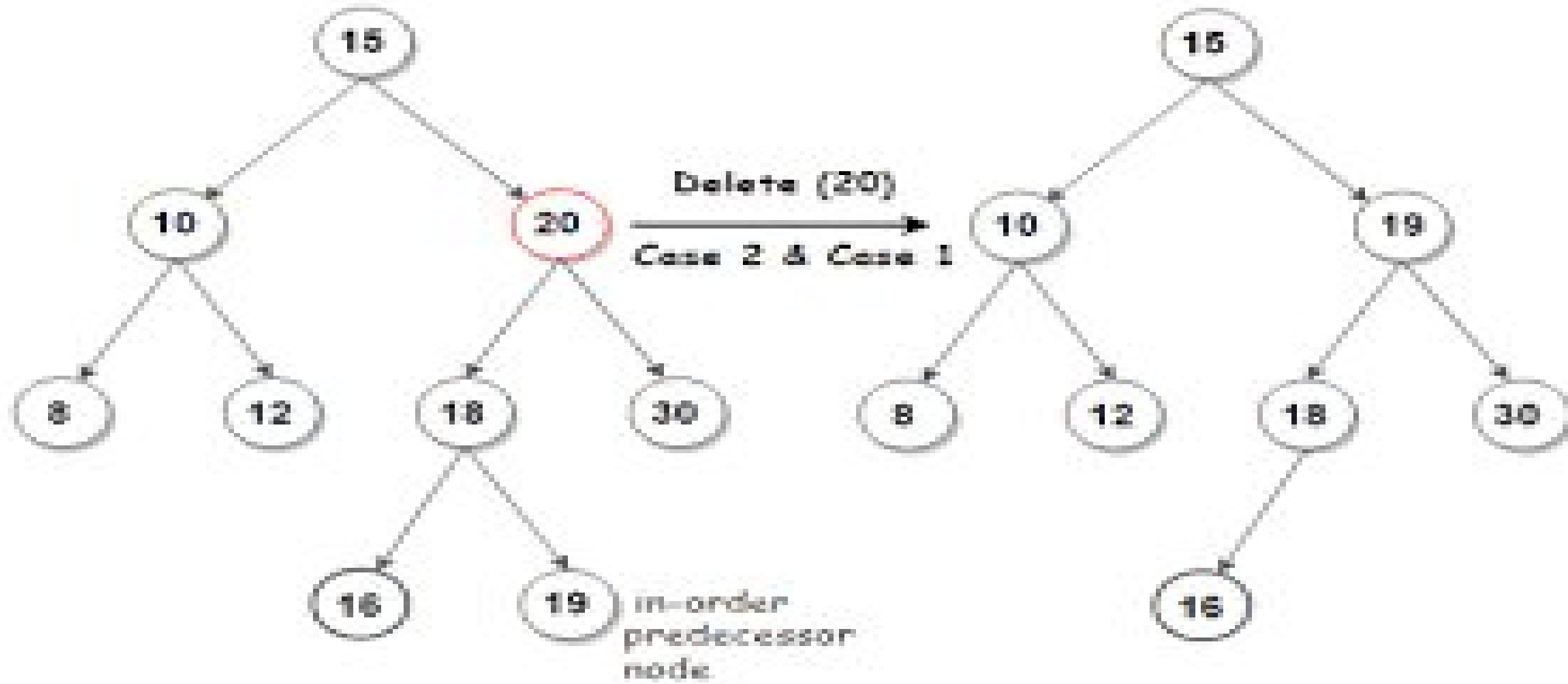


If binary, swap with next item. Now in leaf or unary node; delete.
To find next item, follow lej path from right child.

Delete M ;
Swap with p ;
Delete.



Delete 20



End